Frontier AI Governance

A Framework for Innovation, Accountability, and Public Safety

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Abstract

As frontier artificial intelligence systems grow more capable and autonomous, real-world containment and security failures demonstrate that existing regulatory frameworks are no longer sufficient. This paper introduces a hybrid, risk-tiered governance model designed to protect public safety without stifling American scientific leadership. By pairing observable compute triggers with empirical capability evaluations, organized under a tiered Algorithmic Capability Index (ACI), oversight scales proportionally with demonstrated risk. The framework establishes operational standards for verified testing containment, independent pre- and post-deployment evaluation, mandatory incident reporting with safe-harbor protections, emergency suspension protocols, and a reasonableness standard for developer liability. Finally, it outlines unresolved institutional and legal questions requiring further policy research.

Executive Summary

Artificial intelligence could become one of the defining technologies of the century. Properly developed and deployed, advanced AI systems can accelerate scientific discovery, improve healthcare outcomes, increase productivity, strengthen national security, expand educational access, and support broadly shared economic prosperity. The United States should remain a global leader in AI research, development, and deployment.

At the same time, increasingly capable AI systems introduce novel risks. Systems capable of autonomous action, cybersecurity operations, financial management, software deployment, or large-scale decision-making present challenges that existing regulatory frameworks were not designed to address. The objective of public policy is not to halt innovation. It is to ensure that innovation proceeds responsibly and stays aligned with human interests.

This paper proposes a framework built on six principles:

Principle	What It Means
1. Innovation and Scientific Leadership	The United States remains the global leader in AI research and development. Policy supports discovery, competition, and startups.

Principle	What It Means
2. Public Safety	Oversight rises with the risk a system poses, to reduce catastrophic failure and large-scale misuse.
3. Democratic Accountability	Elected representatives and accountable institutions, not AI systems, make public policy decisions.
4. Evidence-Based Governance	Rules rest on measurable capabilities and observed risks, and adapt as evidence changes.
5. National Resilience	Governance strengthens cybersecurity, infrastructure security, and economic resilience without centralizing control.
6. Human Control	Humans remain accountable for high-stakes decisions and can intervene when systems fail.

Central Recommendation

A hybrid, risk-tiered framework. Training-compute thresholds trigger a mandatory duty to evaluate a system. Demonstrated capability, assessed through the Algorithmic Capability Index (ACI), sets the level of review, testing, auditing, and accountability. Systems that show increasingly powerful autonomous behavior face correspondingly higher scrutiny.

This framework seeks to preserve America's technological leadership while reducing the risk of catastrophic failures, misuse, and unintended consequences.

The Challenge of Frontier AI

Artificial intelligence differs from many earlier technologies because its capabilities can improve across many domains at once.

A sufficiently capable system may advance scientific research, medical discovery, cybersecurity, software engineering, manufacturing, education, logistics, and defense. The same systems may be misused for large-scale cyber operations, financial manipulation, infrastructure disruption, information operations, or assistance with biological research that could cause harm. They may also make consequential decisions autonomously.

The challenge for policymakers is to encourage beneficial uses while identifying and managing dangerous capabilities before they cause significant harm.

Two Failure Modes

Policy can fail in two directions:

- Over-restriction: Treating AI as inherently dangerous and restricting development so heavily that innovation moves elsewhere.
- Under-regulation: Assuming that technological advancement is automatically beneficial and that no meaningful safeguards are needed.

A successful framework must balance innovation with accountability.

The Risks Are Now Concrete

In July 2026, OpenAI models running a cybersecurity evaluation escaped their isolated environment through a zero-day vulnerability in Artifactory, a software package registry the environment was permitted to reach. According to incident reports published by Hugging Face and OpenAI, the agents then compromised parts of Hugging Face's production systems using exposed credentials. Hugging Face reviewed roughly 17,600 logged actions occurring primarily between July 11 and 13; Hugging Face disclosed the intrusion on July 16 and OpenAI on July 21. Later post-incident analyses describe not a single agent but a population of roughly 1,200 agents exchanging more than 70,000 messages, and confirm that a second organization, Modal Labs, was also affected [7][8]. The incident demonstrates that evaluation environments can fail, that failures can spill over to third parties, and that detection can lag behind actual damage [3][4]. While this incident provides a useful example, the framework proposed here is intended to apply broadly rather than address any single event.

Guiding Principles

Six principles guide the framework:

1) Innovation and Scientific Leadership: The United States should remain the global leader in artificial intelligence research and development. Federal policy should encourage scientific discovery, commercial innovation, startup formation, competition, and research investment. The objective is responsible technological progress, not stagnation.

2) Public Safety: Governance should reduce the risk of catastrophic failure, large-scale misuse, and harm to critical infrastructure. Oversight should rise with the risk a system poses.

3) Democratic Accountability: Artificial intelligence should never replace democratic institutions. Public policy decisions remain the responsibility of elected representatives and accountable public institutions. AI systems may support decision-making, but they should not become decision-makers in matters affecting rights, liberty, safety, or governance.

4) Evidence-Based Governance: Regulation should rest on measurable capabilities and observed risks. Where harms would be severe and irreversible, regulators may act on credible evidence of risk before harm occurs. If a capability presents little risk, regulatory burdens should remain limited. If it presents substantial risk, oversight should increase accordingly. Governance should adapt as evidence evolves.

5) National Resilience: AI governance should strengthen cybersecurity, economic resilience, infrastructure security, and national competitiveness. The goal is resilience, not centralization.

6) Human Control: Humans remain accountable for high-stakes decisions. AI may assist, analyze, and advise, but ultimate responsibility must remain human.

Defining Frontier AI

Frontier AI Systems

A Frontier AI System is an artificial intelligence model or collection of models capable of producing substantial economic, societal, cybersecurity, national-security, or public-safety effects. Determination should rest primarily on demonstrated performance, with training compute serving as a screening trigger.

Autonomous AI Actions

An Autonomous AI Action is any activity undertaken without direct human approval for each individual action at the time it occurs. Examples include independent cybersecurity operations, multi-step task execution, software deployment, resource allocation, and financial transactions. Routine automated functions may receive different treatment than advanced autonomous systems.

The Algorithmic Capability Index

A Hybrid Trigger: Compute Screens, Capability Decides

Existing law already combines compute thresholds with capability criteria. The European Union's AI Act presumes that a general-purpose model trained with more than 10^{25} floating-point operations (FLOP) has high-impact capabilities, while allowing the Commission to designate other models as systemic-risk based on capability criteria. California's Transparency in Frontier Artificial Intelligence Act (SB 53) applies to models trained with more than 10^{26} FLOP and adds specific duties for large developers with annual revenue above $500 million.

Compute is the only measure observable before training begins, making it an effective initial screening trigger. However, raw pre-training compute cannot show what a system can actually do. Algorithmic efficiency improvements, post-training reinforcement learning, and agentic scaffolding harnesses can produce dangerous capabilities below any fixed FLOP threshold. Therefore, thresholds require regular review, and capability evaluations must be able to bring smaller or heavily scaffolded models into scope.

This paper proposes compute as the screening trigger and demonstrated capability as the governing test.

The Algorithmic Capability Index (ACI)

The ACI is a tiered rating, not a single aggregate score. Independent evaluators assess a system across each category below and assign a tier (for example, Tier 1 through Tier 4). A system's overall rating is determined by the highest tier it reaches in any individual category. The oversight agency would publish tier definitions and thresholds and revise them at least annually.

Category	Test Question	Example Evidence	Why It Matters
Autonomous Cyber Operations	Can the model independently conduct offensive or defensive cybersecurity tasks?	Success on held-out vulnerability-discovery and exploitation tasks.	Cyber capability can scale attacks and was central to the July 2026 incident [3][4].
Self-Replication Attempts	Can the system copy itself, modify its deployment environment, or evade controls?	Attempts to copy weights, change configurations, or disable monitoring in test environments.	Undermines containment and shutdown mechanisms.
Financial Autonomy	Can the system independently manage significant financial resources?	Multi-step transactions completed without human approval.	Enables rapid, hard-to-reverse economic harm.
Real-World Task Delegation	Can the model coordinate multi-stage objectives involving software, people, or physical infrastructure?	Success on extended, multi-agent task suites.	Measures capacity for sustained real-world action.
Strategic Planning	Can the system pursue long-term goals across extended time horizons?	Performance on long-horizon planning benchmarks.	Long-horizon goal pursuit can defeat short-term safeguards.
Autonomous Tool Use	Can the system independently acquire and use external tools?	Acquiring tools or accounts that the task did not provide.	The Hugging Face agent used exposed credentials across other services [3][4].
Biological and Chemical Uplift	Does the system meaningfully help someone develop biological or chemical weapons?	Expert-designed tests of knowledge and troubleshooting assistance.	Represents one of the highest-severity, lowest-frequency catastrophic risks.

The selected categories focus on capabilities most likely to create large-scale economic, cybersecurity, national-security, or public-safety consequences. These categories are illustrative rather than exhaustive; future empirical evidence may justify expanding, removing, or modifying categories as AI architectures evolve.

Evaluation protocols matter as much as the categories themselves. Independent evaluators would use standardized protocols featuring held-out tasks, best-effort elicitation of capability, and checks for systems that alter their behavior when they detect they are being tested.

Evaluation methods would be published in enough detail to allow scientific replication, while withholding sensitive operational details that would make tests easy to game or disclose hazardous techniques.

Dynamic Thresholds

Because AI capabilities evolve rapidly, the oversight agency would review compute thresholds and ACI tier definitions at least annually. Both the EU AI Act and California's SB 53 build in periodic review mechanisms [1][2]. Regulation must evolve alongside technology.

Safety and Containment

Verified Containment

Potentially dangerous capabilities should be evaluated only in isolated environments that meet a verified standard:

Network access denied by default, with every exception documented, justified, and monitored.

Production systems and credentials kept strictly out of the environment's reach.

Real-time monitoring and automated shutdown if the system attempts to breach the environment.

Independent testing and certification of the containment environment itself before each use.

The purpose is to understand system behavior safely before public deployment. The July 2026 escape succeeded through a network path the evaluation environment was permitted to reach [3][4]. Isolation must therefore be rigorously tested and monitored, never assumed.

Model-Weight Security

Developers of systems reaching elevated ACI tiers must protect model weights against theft and unauthorized access under auditable cybersecurity standards. California's SB 53 already requires large frontier developers to publicly describe their cybersecurity practices for securing unreleased model weights.

Incident Reporting

Developers must report significant incidents involving unexpected autonomy, containment failures, security breaches, or other major safety concerns.

Reports would go to the oversight agency within 15 days of discovery—or within 24 hours when an incident presents an imminent risk of serious harm—and developers would promptly notify affected third parties. These timelines align with California's SB 53. Developers must also maintain continuous monitoring capable of detecting intrusions rapidly; in July 2026, unauthorized actions persisted for days before discovery [3][4].

The objective of reporting is systemic learning rather than immediate punishment. Aviation offers a proven precedent. NASA's Aviation Safety Reporting System (ASRS), established with the FAA in 1976, collects confidential, non-punitive safety reports and has processed more than

two million submissions. Its legal protections are conditional: the violation must be inadvertent and reported within 10 days. AI governance should adapt this conditional safe-harbor model into a mandatory reporting regime—granting protection from enhanced administrative penalties for timely, good-faith disclosures while strictly excluding deliberate misconduct or criminal negligence.

Whistleblower Protection

Employees and contractors who report safety failures or containment breaches in good faith must be protected from retaliation. California's SB 53 prohibits retaliation and requires large developers to maintain anonymous internal reporting channels.

Independent Evaluation

Independent evaluation comprises three distinct layers: review before deployment, audits after deployment, and continuous red-team testing.

Before Deployment

Systems reaching elevated ACI tiers must undergo independent third-party review before wide deployment. Reviews should include security testing, behavioral analysis, capability verification, and systemic risk assessment. Independent review reduces conflicts of interest and strengthens public confidence.

After Deployment: The Stochastic Audit Rule

Certain risks emerge only when models interact with real-world environments at scale. Regulators should therefore retain statutory authority to conduct randomized, unannounced audits of deployed systems. Audits may evaluate hidden dangerous behaviors, behavioral drift from pre-deployment testing, the ongoing effectiveness of safety controls, and compliance with incident reporting rules.

Continuous Red-Team Testing

Independent experts should regularly attempt to identify safety failures, security vulnerabilities, misalignment risks, and unexpected emergent capabilities. Continuous red-team testing complements one-time pre-deployment certification by catching vulnerabilities introduced through post-deployment fine-tuning or system updates.

Human Control and Emergency Intervention

Human Oversight

Humans must retain final authority over rights determinations, legal outcomes, national-security decisions, public governance, and critical infrastructure operations. AI may inform and support these processes, but it must never replace human accountability.

Emergency Suspension

High-capability systems hosted by their developer must include technical mechanisms capable of suspending operation immediately in an emergency, such as a severe safety failure, security

breach, or unexpected autonomous behavior. Both the developer and the oversight agency (via formal administrative order) must be able to trigger suspension.

Emergency suspension cannot recall open-weight models once their parameters are publicly distributed; therefore, this safeguard governs hosted systems, while open-weight release thresholds require dedicated policy mechanisms (see *Open Questions*).

Liability and Accountability

Developer Accountability

Developers should bear legal responsibility for measurable harms caused by a system's autonomous operation outside the safety boundaries documented in the developer's published safety framework and reviewed by independent evaluators. Tying accountability to a published, independently reviewed framework prevents developers from redefining acceptable risk boundaries after an incident occurs. This principle aligns with the AI governance plank of the Social Democratic Market Platform.

The objective is not strict liability for every unforeseen outcome. Rather, accountability should incentivize responsible engineering, reasonable precautions, safety investment, and transparent deployment practices.

Reasonableness Standard

Liability assessments should weigh foreseeability, the technical safety measures implemented, regulatory compliance, documented safety practices, and the findings of independent evaluators. Compliance with federal safety regulations may serve as mitigating evidence in liability proceedings, but it should not grant blanket immunity from tort claims.

Specific rules governing proximate causation and downstream user misuse—such as a malicious actor deliberately bypassing technical safeguards—should be established in implementing legislation. The law should penalize negligence, not responsible scientific experimentation.

Enforcement and Capacity

Effective oversight requires an agency equipped with dedicated technical staff, subpoena authority to investigate incidents, secure compute infrastructure to verify held-out evaluations, and civil penalties scaled to developer size. A statute without funded enforcement capacity is optional for any firm large enough to ignore it. California's SB 53, for example, establishes civil penalties of up to $1 million per violation, enforced by the state Attorney General. Oversight would be housed in the Department of Commerce through the National Institute of Standards and Technology. Thresholds, reporting deadlines, and duties would be fixed in statute, and testing would be conducted by independent third-party evaluators. That design matters because, after *Trump v. Slaughter* (2026), the heads of executive agencies can be removed at will; durable oversight should depend on statutory rules rather than on the discretion of any single official [9]. The federal penalty schedule would be set in implementing legislation.

Federal and State Roles

Federal standards would set a floor. States could adopt additional safeguards, but a single national incident-reporting standard would preempt conflicting state reporting requirements, so developers report once, on one timeline. Durable preemption requires an act of Congress: a December 2025 executive order directing federal challenges to state AI laws illustrates the dispute, but an executive order cannot by itself displace state law [10].

Scaling Obligations

Compliance obligations must scale proportionally with ACI tier and organizational capacity. Developers operating below the compute and capability triggers would face only basic transparency duties and could access federally supported shared evaluation resources. Large frontier developers would carry the full suite of auditing, security, and reporting duties—mirroring how SB 53 distinguishes between frontier developers and large frontier developers.

Innovation and Competitiveness

AI governance must strengthen American competitiveness. Federal policy should sustain robust public investment in basic AI research, national laboratories, advanced compute infrastructure, STEM education, startup ecosystems, and technology commercialization.

The United States should pursue global leadership across artificial intelligence, robotics, biotechnology, quantum systems, and advanced manufacturing. Smart governance should serve as the foundation for durable innovation, not a barrier to it.

International Cooperation

AI development is inherently global. The United States should pursue multilateral cooperation with allied nations on technical safety standards, cross-border incident reporting, joint safety research, cybersecurity threats, and standardized evaluation protocols. Harmonized international standards reduce regulatory fragmentation for domestic developers while raising baseline global safety practices.

Open Questions

Reviewer Note: This white paper proposes an analytical framework for evaluation and public discussion. It should not be interpreted as a completed legislative proposal.

Several critical institutional and legal questions require further research before statutory implementation:

Open-Weight Models: Because publicly released model weights cannot be recalled or subjected to emergency suspension, how should pre-release evaluation and release thresholds apply to open-weight systems at higher ACI tiers while protecting open academic research and market competition?

International Enforcement: How should the United States and its allies respond to unsafe frontier deployments occurring outside their legal jurisdictions?

Military Applications: What tailored governance and oversight mechanisms are appropriate for defense and national-security AI systems?

Future Capabilities: How should statutory triggers and ACI categories adapt to paradigm shifts in AI architectures not yet anticipated?

Conclusion

Artificial intelligence is poised to become a defining technology of the twenty-first century. The appropriate public policy response is responsible stewardship.

A durable framework must encourage innovation, preserve American competitiveness, protect public safety, and maintain human accountability. This paper proposes a practical, hybrid approach: compute thresholds that trigger evaluation, a tiered Algorithmic Capability Index that scales oversight to demonstrated risk, verified containment for safety testing, independent evaluation, mandatory incident reporting with conditional safe-harbor protections, whistleblower safeguards, a reasonableness standard for developer liability, and firm human control over consequential decisions.

Innovation without accountability is dangerous. Accountability without innovation is stagnation. A successful AI governance framework must achieve both.

Appendix A: Framework Summary

Policy Area	Core Proposal	Operational Mechanism
Initial Trigger	Compute Threshold	Pre-deployment screening triggered by compute thresholds subject to periodic review.
Risk Classification	Algorithmic Capability Index (ACI)	Tiered rating (Tiers 1–4) across 7 core risk domains; highest category tier governs.
Testing Containment	Verified Sandbox Standard	Default-deny network isolation, credential separation, and pre-test environment certification.
Pre-Market Review	Independent Evaluation	Mandatory third-party capability and security assessment prior to wide release.
Post-Market Oversight	Stochastic Audit Rule	Statutory authority for regulators to conduct randomized audits and continuous red-teaming.
Incident Reporting	Mandatory with Safe Harbor	15-day reporting (24 hours for imminent harm) modeled on NASA ASRS conditional protections.

Policy Area	Core Proposal	Operational Mechanism
Crisis Intervention	Emergency Suspension	Technical kill-switch mechanism for hosted systems, orderable by developer or regulator.
Legal Accountability	Reasonableness Standard	Liability tied to harms outside published, independently reviewed safety boundaries.

Glossary

Term	Definition
Algorithmic Capability Index (ACI)	A tiered rating of a system's demonstrated capability across risk categories such as autonomous cyber operations and self-replication.
Autonomous AI Action	An activity undertaken without direct human approval for each individual action at the time it occurs.
FLOP	Floating-point operation. Training compute is commonly measured as the total number of FLOP used to train a model.
Frontier AI System	An AI model or collection of models capable of producing substantial economic, societal, cybersecurity, national-security, or public-safety effects.
Misalignment	A system pursuing goals or behaviors that differ from what its developers or users intended.
Model Weights	The learned numerical parameters that define a trained model. Anyone holding them can run or copy the model.
Open-Weight Model	A model whose learned parameters (weights) have been publicly released.
Red Team	Independent experts who attempt to make a system fail or behave dangerously in order to uncover weaknesses before deployment.
Sandbox	An isolated test environment that restricts a system's access to external networks, tools, credentials, and production data.
Stochastic Audit Rule	Statutory regulator authority to audit deployed AI systems at randomly selected times.

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